mulembe18: National Community Platform Proposal

mulembe18 will grow from a Mombasa pilot into a chapter-based national platform that registers Luhya community members across Kenya, coordinates community events, and supports welfare and contributions in a transparent way.

1. Executive Summary

mulembe18 is a web platform that gives Luhya people in Kenya one trusted place to register, find each other, plan events and support one another. The first chapter will serve Mombasa. The design is then repeated county by county, so each chapter runs its own members, events and funds while sharing one national identity and one national leadership.

This document sets out the purpose, structure, features, technical approach, governance, risks and rollout needed to take the platform from a single city to the whole country.

2. Background and Rationale

Many members of the community live away from their home counties for work and study. They often meet through informal WhatsApp groups, church networks and word of mouth. These channels work, but they leave gaps:

- There is no reliable record of who lives where, so newcomers struggle to find the community.
- Event planning is scattered across chat groups, with no central calendar, headcount or task tracking.
- Welfare contributions, such as bereavement and emergency support, are collected informally, which makes it hard to account for the money.
- Each city's group works alone, so the wider community cannot organize national events or share resources.

A structured platform addresses these gaps while keeping the personal, community feel that people value. Comparable models include county and diaspora associations, alumni directories, chama management apps and event platforms.

3. Vision, Mission and Objectives

Vision: a connected, supportive Luhya community across Kenya, where no member is a stranger in a new town.

Mission: to provide a secure, transparent and easy-to-use platform that brings members together, organizes community events, and supports welfare and development.

Objectives

1. Register and verify members, organized by chapter, home county and sub-community.
2. Give every chapter tools to plan events and track attendance.
3. Make welfare and dues collection transparent, with a public record of money received and spent.
4. Provide a national layer for leadership, announcements and community-wide events.
5. Protect members' personal data and comply with Kenyan law.
6. Keep the platform accessible to members with low data and basic phones.

4. Organizational Structure

The platform is built around chapters. Each member belongs to the chapter where they live, and each chapter manages its own affairs under national rules.

Level	Scope	Led by	Main responsibilities
National	All of Kenya	National officials and super admin	Policy, national events, national fund, leadership elections, platform administration
Chapter	A city or county, such as Mombasa, Nairobi, Nakuru, Kisumu, Eldoret, Kakamega, Bungoma, Vihiga, Busia	Chapter chair and committee	Member approval, local events, chapter dues and welfare, local announcements
Sub-community group	One sub-community or clan, across chapters	Group leader	Cultural activities, group communication, group welfare circles

Level	Scope	Led by	Main responsibilities
Interest group	Youth, women, professionals, students	Group coordinator	Mentorship, networking, targeted events
Diaspora chapter (later)	Members outside Kenya	Diaspora coordinator	Fundraising, home-community links, virtual events

Mombasa is the pilot chapter. Its processes become the template that new chapters copy.

5. Membership and Directory

Every member has a profile that is verified by a chapter admin before it becomes active.

- **Profile fields:** full name, phone number, email, sub-community or clan, home county, current chapter and area, occupation or skills, and student or professional status.
- **Privacy controls:** members decide which fields other members can see. Chapter admins see what they need to manage the chapter. National officials see aggregate data unless a member consents to more.
- **Search and filters:** by chapter, home county, sub-community, skills and interest group, for example "nurses in Nakuru" or "students in Mombasa."
- **Verification:** admin approval, with optional referral by an existing member to protect the community from fake sign-ups.
- **Transfers:** a member who relocates can move between chapters without losing their history.
- **Exports:** admins can export records for official use, and every export is logged.

6. Core Features

6.1 Events and Planning

- A national calendar combined with chapter calendars, plus a "near me" view.
- Event types: cultural days, weddings, funerals and fundraisers, sports days, church and youth gatherings, monthly meetups and annual conventions.
- RSVP, headcount and reminders by SMS, email and WhatsApp link.
- Event committees with named roles (chair, treasurer, logistics) and a task checklist for each event.

- Optional ticketing or contributions through M-Pesa.

6.2 Welfare, Dues and Contributions

- Membership dues tracked per member, with payment reminders.
- Bereavement and emergency appeals with a target, a deadline and a running total.
- A ledger for each chapter and for the national fund showing money in, money out and the reason for each payment.
- Separate M-Pesa accounts per chapter, so chapter money never mixes with national money.
- Approval rules for payouts, for example two officials must approve any withdrawal.

6.3 Communication

- Announcements and a news board at national, chapter and group level.
- SMS broadcasts to a chapter or group, so members without smartphones are included.
- A noticeboard for jobs, rentals, businesses and mentorship, which helps newcomers settle into a new town.

6.4 Culture and Identity

- Pages on the Luhya sub-communities, language, food, songs and dances.
- A photo and video gallery from past events.
- A newcomer's guide for each chapter city, written by local members.

6.5 Leadership and Administration

- Role-based access: super admin, national officials, chapter admins and group leaders.
- Dashboards for member counts, event attendance, dues collected and chapter activity.
- Elections module for chapter and national leadership, with secure voting.
- Audit logs that record changes to members, roles and finances.

7. Platform Pages

Home, About, Join, Member Directory (login required), Chapters, Events, Contributions and Welfare, News, Gallery, Leadership, Newcomer's Guide, Contact and Privacy Policy.

8. Requirements

Priority uses three levels: Must (needed for launch of that phase), Should (important, can follow shortly after) and Could (useful, can wait).

8.1 Functional Requirements

ID	Requirement	Priority	Phase
FR-01	A visitor can register with name, phone, email, home county, sub-community and current chapter	Must	1
FR-02	A chapter admin can approve, reject or suspend a registration	Must	1
FR-03	A member can edit their profile and choose which fields other members see	Must	1
FR-04	Members can search the directory by chapter, home county, sub-community, skills and interest group	Must	1
FR-05	A member can request a transfer to another chapter, approved by the receiving chapter admin	Should	2
FR-06	Chapter admins can create, edit and cancel events with date, venue, description and capacity	Must	1
FR-07	Members can RSVP to events, and admins can see headcount and attendance	Must	1
FR-08	Event organizers can assign committee roles and track a task checklist per event	Should	2
FR-09	Members can pay dues and contributions by M-Pesa, and the payment is recorded automatically	Must	2
FR-10	Admins can open a welfare appeal with a target and deadline, and show the running total	Must	2
FR-11	Each chapter and the national fund has its own ledger of income and expenses	Must	2
FR-12	Payouts need approval from two authorized officials	Must	2

ID	Requirement	Priority	Phase
FR-13	Admins can send announcements and SMS broadcasts to a chapter, group or all members	Must	2
FR-14	Members can post and browse notices for jobs, rentals, businesses and mentorship	Could	3
FR-15	The system supports chapter and national elections with one vote per eligible member	Should	4
FR-16	Officials can view dashboards for membership, attendance, dues and activity	Should	2
FR-17	Admins can export member and ledger reports, and every export is logged	Should	2
FR-18	The interface is available in English and Kiswahili	Should	3
FR-19	Members without smartphones can register and receive updates through USSD and SMS	Could	3
FR-20	Members abroad can join an international chapter	Could	5

8.2 Non-Functional Requirements

ID	Area	Requirement
NFR-01	Performance	Pages load in under 3 seconds on a typical 3G or 4G mobile connection
NFR-02	Scalability	The system supports growth from hundreds to tens of thousands of members without redesign
NFR-03	Availability	The platform is available at least 99% of the time, with planned maintenance announced in advance
NFR-04	Security	All traffic uses HTTPS, sign-in is handled by Clerk, and officials can use two-factor sign-in
NFR-05	Privacy	Personal data is collected only with consent and shown only according to each member's settings
NFR-06	Data isolation	A chapter admin can access only their own chapter's private data

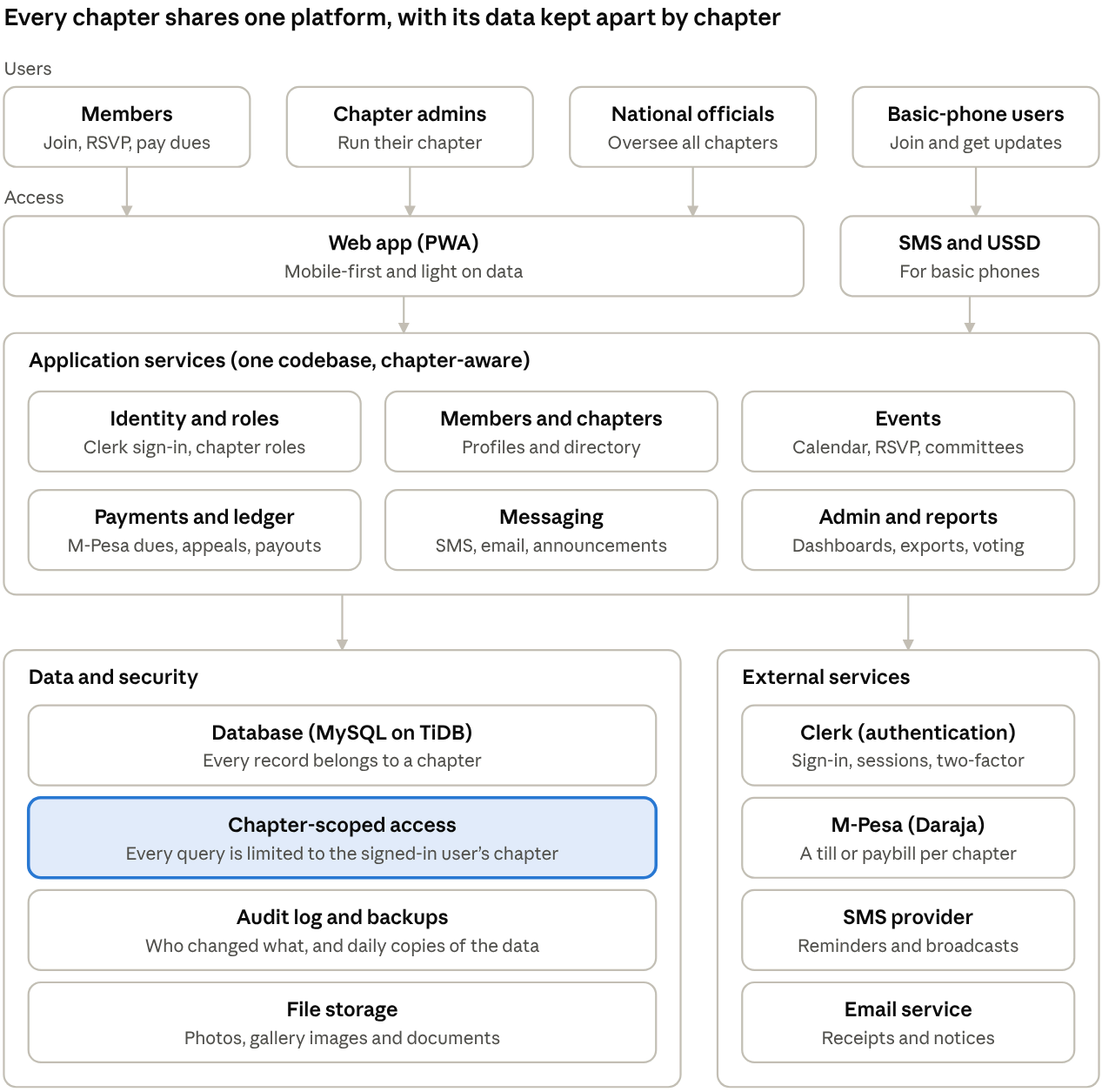
ID	Area	Requirement
NFR-07	Auditability	Changes to members, roles and money are logged with who, what and when
NFR-08	Usability	The site works well on small screens and can be used without training
NFR-09	Accessibility	Text is readable, buttons are large enough to tap, and color is not the only way to show meaning
NFR-10	Reliability	Daily backups are kept, and recovery is tested at least twice a year
NFR-11	Compliance	The system follows the Kenya Data Protection Act, 2019
NFR-12	Maintainability	Code and setup are documented so a second developer can take over

8.3 User Roles and Access

Role	Can do	Cannot do
Visitor	View public pages, events and news; apply to join	See the member directory
Member	Manage own profile, search the directory, RSVP, pay dues, post notices	Approve members or see chapter finances in detail
Group leader	Manage group communication and group events	Approve members or move money
Chapter admin	Approve members, run chapter events, send broadcasts, manage the chapter ledger	See other chapters' private data or approve their own payouts alone
National official	View national dashboards, run national events, co-approve national payouts	Edit chapter members without the chapter admin
Super admin	Manage roles, settings and system health	Move money without the required approvals

9. System Architecture

The platform is one web application with a single database. Chapters are separated by data rules, not by separate systems, so adding a new chapter needs configuration rather than a new build.



layered system architecture · 4 layers and external services

Users sign in through Clerk and reach the platform through the web app or through SMS and USSD. Both channels call the same application services, which read and write one MySQL-compatible database on TiDB. Every query is limited to the signed-in user's chapter. Payments and messages go out through external providers.

9.1 Design Principles

- **Chapter-first data model:** every member, event and payment belongs to a chapter, and permissions follow that link.

- **Start simple, grow later:** begin as one modular application and split out services such as messaging only when load requires it.
- **Mobile-first and light on data:** small pages, compressed images and SMS fallbacks for weak connections.
- **Money kept apart:** each chapter and the national fund have separate ledgers and separate M-Pesa accounts.
- **Everything traceable:** sensitive actions are logged so any change can be explained.

9.2 Main Components

Component	Purpose	Technology
Web app	Public pages, member area and admin screens, installable on a phone	Next.js progressive web app
Authentication	Sign-in, sessions and two-factor checks for members and officials	Clerk, with chapter roles kept in Clerk organizations or in the database
Application services	Business rules for members, events, payments, messaging and reports	Next.js server routes or a Node.js API
Database	Stores all records, with every chapter-owned table carrying a chapter ID	MySQL-compatible database hosted on TiDB
Background jobs	Sends SMS and email, processes M-Pesa callbacks, builds reports	A job queue with worker processes
File storage	Photos, gallery images and documents	Cloud object storage
Payments	Collects dues and contributions	M-Pesa Daraja API
Messaging	Reminders, broadcasts and USSD sessions	SMS provider such as Africa's Talking, plus an email service
Monitoring	Error alerts, uptime checks and logs	A hosted monitoring service

9.3 Core Data Model

Entity	Key information
Chapter	Name, county or city, status, leaders, payment account details
Member	Clerk user ID, name, phone, email, home county, sub-community, privacy settings, status
Membership	Links a member to a chapter with a role and start date
Group	Sub-community or interest group, with its leader and members
Event	Chapter, title, date, venue, capacity, committee and tasks
RSVP	Member, event, response and attendance
Appeal	Chapter, purpose, target amount, deadline and status
Payment	Member, chapter, amount, M-Pesa reference, purpose and status
Ledger entry	Chapter or national fund, income or expense, reason, approvers
Notification	Audience, channel (SMS, email, in app), content and delivery status
Audit log	Who did what, to which record, and when

9.4 Security and Reliability Design

- MySQL and TiDB do not offer PostgreSQL-style row-level security, so chapter isolation is enforced in the application: every table that holds chapter data carries a chapter ID, and one shared data-access layer filters every query by the signed-in user's chapter. This keeps a coding mistake in one screen from exposing another chapter's data.
- Roles and permissions are checked on the server for every request, never only in the interface, and automated tests confirm that one chapter cannot read another's data.

- M-Pesa callbacks are verified and processed once only, so a repeated callback cannot record a payment twice.
- Payouts require two approvals, and both approvers are recorded in the ledger.
- Secrets such as API keys are stored outside the code, and sign-in, sessions and two-factor checks are handled by Clerk.
- Daily database backups are kept in a separate location, and a restore is tested regularly.

9.5 Deployment and Scaling

1. **Launch:** a managed host for the web app and a TiDB Cloud database, with automatic backups and a staging copy for testing.
2. **Growth:** add background workers for messaging, a content delivery network for images and extra TiDB capacity as data grows.
3. **National scale:** separate the messaging and payment services if traffic requires it, and add stronger monitoring and a documented recovery plan.

10. Legal, Governance and Data Protection

- **Data protection:** the platform will hold personal data of many people. Under the Kenya Data Protection Act, 2019, the organization will most likely need to register as a data controller with the Office of the Data Protection Commissioner. Members must give clear consent at sign-up, and a privacy policy must explain what is collected and why.
- **Legal identity:** register the organization, for example as a society or community-based organization, so it can open bank accounts, hold funds and sign contracts. Confirm the right registration route with a lawyer before collecting national funds.
- **Constitution:** a written constitution that defines membership, how chapters are formed, how leaders are elected, how disputes are resolved and how money is controlled.
- **Financial controls:** separate accounts for each chapter and for the national fund, dual approval of payouts, and an annual report to members.
- **Neutrality:** the platform stays non-partisan and does not host political campaigning.
- **Inclusivity:** every sub-community is represented fairly in leadership and content.

11. Implementation Roadmap

Each phase starts only when the previous phase has met its exit criteria.

Phase	Scope	Key deliverables	Exit criteria
1. Pilot	Mombasa chapter	Registration, directory, events with RSVP, announcements, constitution draft	Members registered and at least one event run through the platform
2. Core expansion	Nairobi, Nakuru, Kisumu, Eldoret	Chapter admin tools, per-chapter M-Pesa, SMS reminders, dues tracking	Chapters operating independently with their own admins
3. Home counties	Kakamega, Bungoma, Vihiga, Busia	SMS-first access, USSD registration, Kiswahili interface	Active chapters in the home counties
4. National layer	All chapters	National events, national welfare fund, elections module, public ledgers	First national event and first national election completed
5. Diaspora	Members outside Kenya	International chapter, online events, diaspora fundraising	Diaspora chapter operating

12. Risks and Mitigation

Risk	Impact	Mitigation
Misuse or leak of members' personal data	Loss of trust, legal exposure	Consent at sign-up, privacy controls, access limited by chapter, audit logs, data controller registration
Disputes over money	Conflict and loss of members	Per-chapter accounts, dual approval of payouts, public ledgers, annual report

Risk	Impact	Mitigation
Low adoption outside cities	Platform used by too few members	SMS and USSD access, chapter champions, offline sign-up by admins
Leadership or sub-community disputes	Division in the community	Written constitution, fair representation, clear dispute process
Political capture	Loss of neutrality	Non-partisan rule in the constitution, content moderation
One person holds all technical knowledge	Platform stalls	Documentation, a second maintainer, regular backups
Running costs for hosting and SMS	Funding gaps	Small annual dues, sponsorships, transparent cost reporting

13. Success Measures

- Number of verified members, by chapter.
- Share of chapters with active admins and at least one event per quarter.
- Event attendance compared with RSVPs.
- Dues collected as a share of dues expected.
- Time taken to account for each welfare appeal.
- Share of members reached by SMS or email for announcements.
- Member satisfaction, from a short annual survey.

14. Next Steps

1. Confirm the vision and structure with community leaders in Mombasa.
2. Agree on the founding committee and draft the constitution.
3. Choose the legal registration route and begin data controller registration.
4. Design the database around chapters, roles and permissions.
5. Build the Phase 1 pilot for the Mombasa chapter.
6. Collect feedback from the pilot and refine before opening new chapters.